1) 3 layer fully connected network dimensions → 784 → 256 → 128 → 10

So lets assume initial neurons are just inputs so 784 inputs

MAC for 1st layer = 256 × 784 = 200704

MAC for 2nd layer = 128 × 256 = 32768

MAC for 3rd layer = 10 × 128 = 1280.

No of MAC operations = output dimensions (or) no of neurons × input dimension (or) no of input neurons

2) Total MACs for one forward pass = 1st layer mac operation + 2nd layer MAC operation + 3rd layer MAC operations

Total MAC operations = 200704 + 32768 + 1280 = 234,752

3) The total number of trainable parameters (weights only, no biases).

= 256 × 784 + 256 × 128 + 128 × 10

= 234,752 parameters.

4) weight memory = Total parameters × register size (FP32 = 4 bytes)

= 234,752 × 4 bytes = 939008 bytes = 917 KB

5) Total Activation memory = 784 × 4 + 256 × 4 + 128 × 4 + 10 × 4

= 3136 + 1024 + 512 + 40 = 4712 Bytes.

≈ 4.6 KB

6) Arthematic intensity = $\frac{2 \times \text{total MAC}}{(\text{weight bytes} + \text{active bytes})} = \frac{2 \times 234752}{(939008 + 4712)} = \frac{469504}{943720}$

= 0.497503496799 8 ≈ 0.497 FLOP/Byte.

~~The workload is memory bound~~